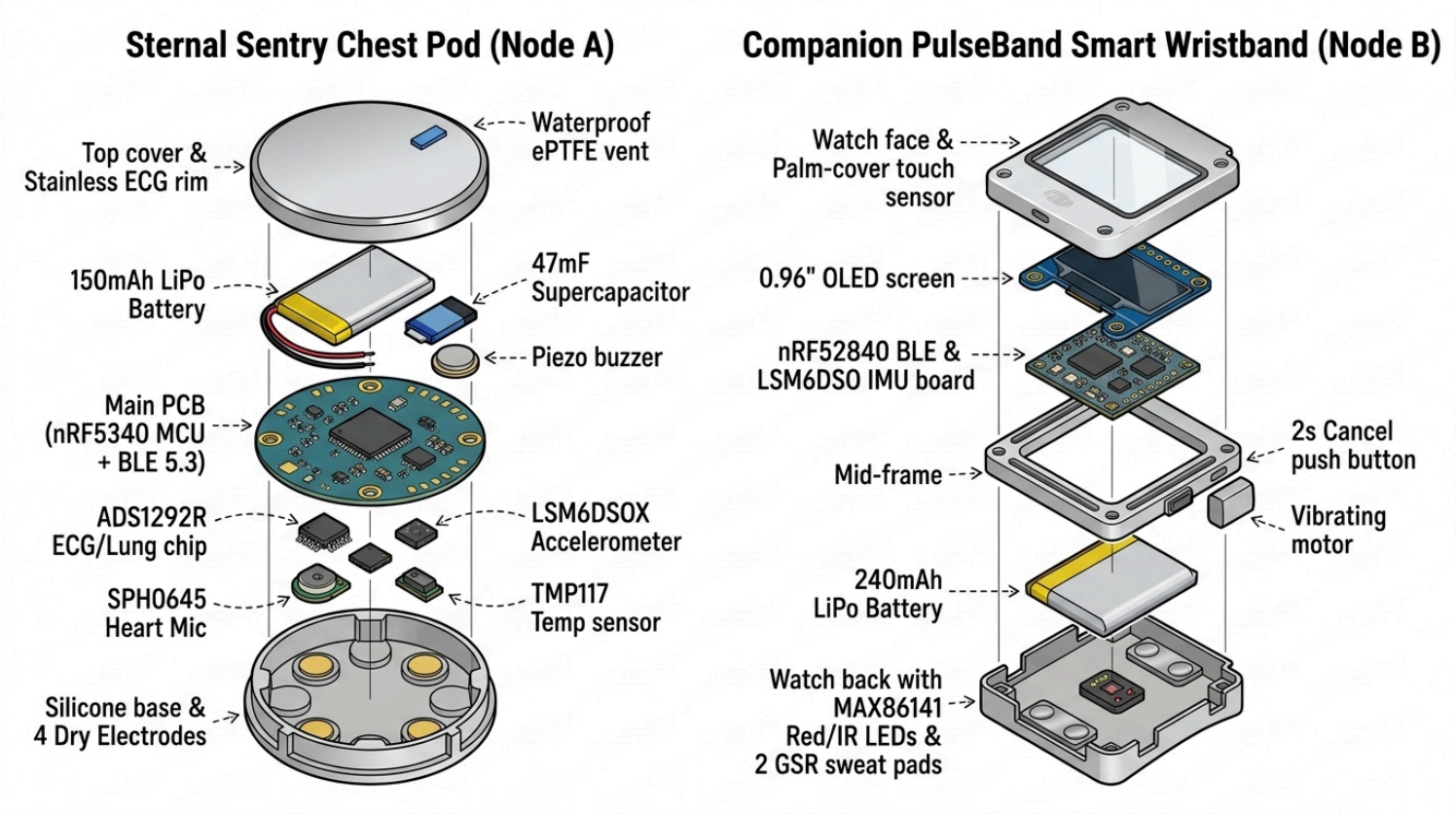
SHEET 1 / 3 — 3D EXPLODED PHYSICAL CASING & ASSEMBLY STACK-UP (NODE A & NODE B)

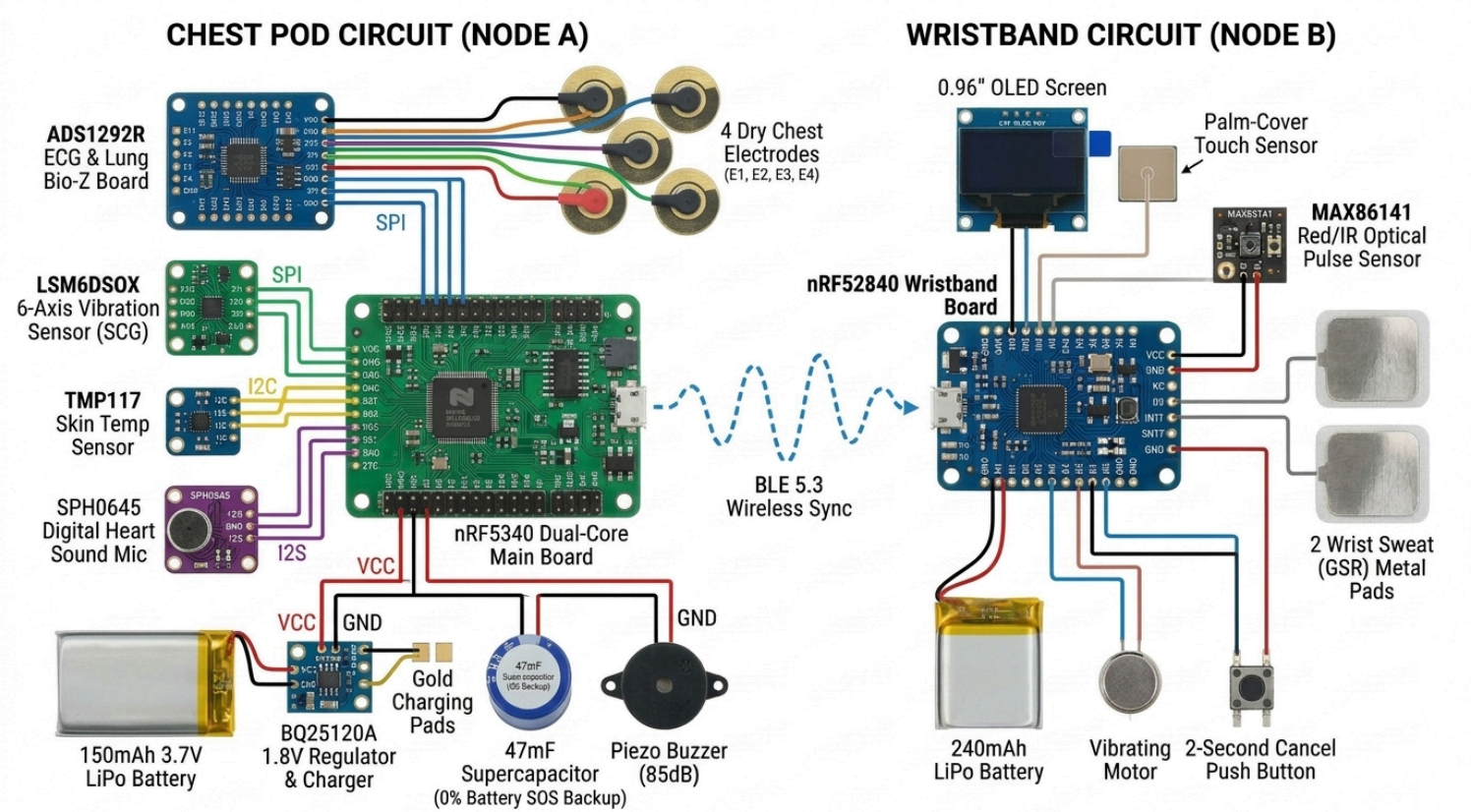
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Physical Layer / Module	Dimensions & Z	Exact Hardware Components & Part Numbers	Why It Goes Here (Biophysical & Mechanical Verification)
1. Top Cover & Vent (Node A)	`Z = 5.80–6.80 mm` (1.00 mm)	Polycarbonate + LSR cap, **316L Stainless Spot-ECG Rim**, **Gore GAW333 Oleophobic ePTFE Vent** (0.20 µm pore), 35% compressed LSR gasket.	Equalizes **8.53 kPa hot-shower cooling vacuum** while resisting soapy water (>300 kPa); finger touch on 316L rim completes Lead I Spot-ECG.
2. Battery & SOS Bay (Node A)	`Z = 4.20–5.80 mm` (1.60 mm)	**150 mAh LiPo NMC (3.2g)** in stepped PCB cutout + 0.40mm swell gap, **Murata 47 mF Supercap** (5.5V, 40mΩ), **PKLCS1212 85dB Piezo**.	Stepped cutout drops Center of Gravity to **h_CoG = 3.10 mm (-81.3% walking tilt torque)**; 47mF supercap gives **10.18× energy margin** at 0% battery.
3. 4-Layer Rigid-Flex PCB	`Z = 3.00–4.20 mm` (1.20 mm)	**Nordic nRF5340-QKAA** (7×7mm Dual Cortex-M33), **W25Q128 16MB QSPI**, **TI BQ25120A** PMIC, **LTC3226** Supervisor, Dual 2-oz Cu planes.	1.50 mm split-ground moat isolates digital/RF noise from microvolt ECG; dual 2-oz (70 µm) copper planes keep steady-state skin rise to **+0.022°C (<41°C)**.
4. Bottom Sensor Cavity	`Z = 0.80–3.00 mm` (2.20 mm)	**ST LSM6DSOX** @(0,0) epoxied to rigid bone pillar, **TI ADS1292R** (24-bit ECG + Bio-Z), **Knowles SPH0645** I2S Mic + Poron seal, **TI TMP117**.	**Zero-Foam Rule** at (0,0) achieves critical Kelvin-Voigt damping (**f_n = 19.63 Hz, ζ = 0.707**) across the 18–35 Hz Aortic Opening (AO) valve band.
5. Silicone Base & Wristband	`Z = 0.00–0.80 mm` +	R=85mm Shore A 30 LSR base w/ **4 Dry Ag/AgCl/TiN**	55 mm tetrapolar span bypasses >500 kΩ dead skin (**106 dB CMRR**); 45

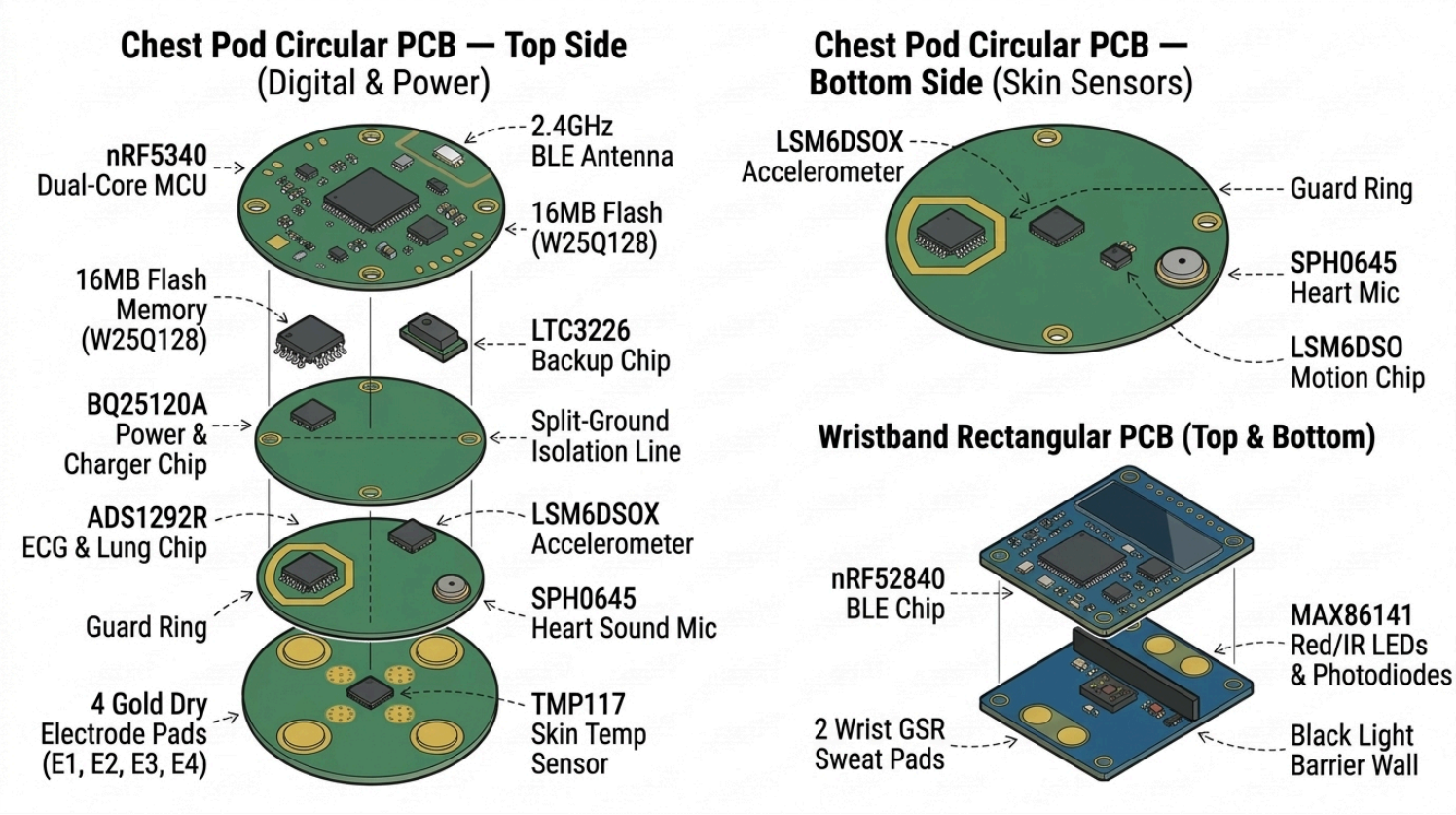
SHEET 2 / 3 — PHYSICAL CIRCUIT & COMPONENT WIRING DIAGRAM ("HOW EVERYTHING CONNECTS")

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Hardware Bus / Wire	Master MCU Pins (nRF5340 / nRF52840)	Connected Sensor / Module Pins	Signal Protocol, Sampling Rate & Protection Circuitry
Blue Wires: SPIM1 (ECG & Lung Bio-Z)	P0.19 (SCLK), P0.20 (MOSI), P0.21 (MISO), P0.22 (/CS), P0.05 (/DRDY IRQ)	**TI ADS1292R** Pins 18–22 (SPI), Pins 3/4 (E2/E3 ECG), Pins 30/31 (E1/E4 32kHz Bio-Z), Pin 2 (RLD)	**4 MHz Isolated SPI**; 500 Hz 24-bit ECG + 32 kHz 4-pad lung impedance; 22.1 kΩ + TVS diodes (<10 µA fault); Murata FB1 (1.8 kΩ) cuts AVDD ripple to **0.04 mV**.
Green Wires: SPIM0 (Chest Vibration SCG)	P0.08 (SCLK), P0.09 (MOSI), P0.10 (MISO), P0.11 (/CS), P0.04 (INT1 Wake)	**ST LSM6DSOX** Pin 13 (SCL), Pin 14 (SDI), Pin 12 (SDO/CS), Pin 4 (INT1 9KB FIFO Watermark)	**8 MHz DMA SPI**; buffers 104 Hz 3-axis SCG into 9 KB FIFO while MCU sleeps at 3.2 µA; wakes every 1.50 s for **28 µs NLMS+DWT** & **1.8 ms TCN-Mamba AI**.
Purple Wires: I2S0 (Heart Sound Mic)	P0.25 (BCLK), P0.26 (LRCLK), P0.27 (SDIN), P0.28 (MIC_PWR_EN Gate)	**Knowles SPH0645LM4H-B** Pin 4 (BCLK), Pin 2 (LRCLK), Pin 1 (DOUT), Pin 6 (VDD)	**1.024 MHz Direct Digital Audio** (24-bit PCM decimated to 1 kHz, 15–40 Hz passband); event-gated in Stage 2/3 to confirm S4/S3 ventricular stiffness gallops.
Yellow Wires: TWIM0 (I2C Temp & Power)	P0.02 (SCL), P0.03 (SDA), P0.06 (TMP_ALERT), P0.29 (TPS22918 Reset)	**TI TMP117** (0×48), **TI BQ25120A** (0×6A PMIC), **TI BQ27426** (0×55 Gauge), **TPS22918**	**400 kHz I2C**; hardware thermal cutout at 40.5°C (<41.0°C IEC 60601-1); automatic **9-clock SCL bit-bang + 50 ms power reset** recovers hung sensors.
Red/Black + BLE 5.3 (SOS &	P0.00 (NMI ← LTC3226 /PFO @ 3.05V);	**Murata 47mF Supercap** + 85dB Piezo; Wrist **MAX86141**	**<80 µs BLE 5.3 Anchor Sync** + 1 Hz (16B) wrist token (cuts radio power

SHEET 3 / 3 — 3D EXPLODED CUSTOM 4-LAYER PCB STACK, ISOLATION ZONES & CHIP PLACEMENT



PCB Layer / Zone	Copper / Coordinate	Populated ICs & Passive Structures	EMC, Signal Integrity & Thermal Verification Rule
Chest PCB — Top Side (Digital, RF & Power)	Layer 1 (1-oz / 35 µm Cu) North of `Y = +3.8 mm`	**U1: nRF5340-QKAA** @(0, +7.5mm), **ANT1: 2450AT18B100** @(0, +14.8mm), **U2: W25Q128** @(-7.5, +7.5mm), **U7: BQ25120A**, **U9: LTC3226**.	6.0×4.0 mm 4-layer copper keep-out under ANT1 prevents chest tissue detuning; stepped PCB window (26×21 mm) recesses 150 mAh LiPo & 47 mF supercap.
Internal Planes L2 & L3 (Split-Ground & Heat)	Layers 2 & 3 (Dual **2-oz / 70 µm Cu**, 385 W/m·K)	**1.50 mm Non-Conductive Split-Ground Moat** between North `DGND` and South `AGND`; single Star-Ground bridge under `U4 Pin 20`.	Prevents 128 MHz MCU & 2.4 GHz BLE return currents from entering the analog ECG plane; spreads 4.23 mW laterally (R_{th} = 4.2°C/W → **+0.022°C skin rise**).
Chest PCB — Bottom Side (Skin-Contact Sensors)	Layer 4 (1-oz / 35 µm Cu) Center & South Zone	**U3: LSM6DSOX** @(0.0, 0.0mm), **U4: ADS1292R** @(-9.5, -2.0mm) inside **Active RLDOUT Guard Ring**, **U5: SPH0645** @(+8.5, -2.0mm), **U6: TMP117**.	Dead-center (0,0) placement of U3 eliminates rotational pitch/roll lever-arm cross-talk; Active RLD Guard Ring achieves **106.0 dB dry-electrode CMRR**.
Outer Rim Electrodes & Charging Contacts	`R = 14.5 mm` Perimeter (55.0 mm Span)	**4 Gold Dry Electrode Feedthroughs (E1–E4)** + 22.1 kΩ + TPD4E05U06 TVS ESD arrays; **2 Schottky-Isolated Gold Pogo Pads**.	Survives ±8 kV contact ESD; reverse Schottky + P-MOSFET gate cuts wet-skin pogo leakage to **<10 nA (ΔV = 0.00 mV)**, preventing sweat electrolysis burns.
Wristband Rectangular PCB (Top	`34.0 × 18.0 × 0.80 mm` 4-	Top: **U11: nRF52840** + **U12: LSM6DSO**; Bottom: **U15:**	12.0 mm opaque black LSR wall blocks skin-shunted light (>60 dB